



SENG 31242 – System Design Project Technology Stack Justification Report

Group 10

SE/2022/046 - Lochithya Hettiarachchi

SE/2022/005 - Anushka Dilinuwan

SE/2022/010 - Adhithya Jayawardhana

SE/2022/025 - Vindya Wickramathilaka

SE/2022/041 - Ramesh Krishan

Supervisor – Eng. Dr. Tiroshan Madushanka

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Technology Stack Justification Report– SithCare Channeling App

1. Introduction

SithCare is a web-based mental health care platform designed to provide digital mental health support services including therapist appointment booking, telemedicine, emergency support, anonymous community consultation, volunteer listener support, and AI-assisted triage.

The selected technology stack and architectural decisions were chosen to ensure scalability, maintainability, security, usability, and rapid development within the project timeline.

2. Core Technology Stack: MERN

The platform will be built using the MERN stack (MongoDB, Express.js, React.js, Node.js). This full-stack JavaScript environment allows the team to maintain high development velocity using a unified language ecosystem, fulfilling the 8-week MVP design and development timeline.

2.1 Frontend Technologies

1. React + Vite

- Purpose: Frontend web application development.

- Justification:

- Component-based architecture improves maintainability.
- Fast rendering and reusable UI components.
- Efficient state management.
- Suitable for scalable single-page applications.
- Large community and ecosystem support.

2. Tailwind CSS

- Purpose: Responsive UI styling.

- Justification:

- Rapid UI development.
- Utility-first responsive design.
- Consistent design system.
- Simplifies glass-morphism UI implementation.
- Lightweight production builds.



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2. Backend Technologies

1. Node.js + Express.js

- Purpose: REST API backend development.

- Justification:

- Lightweight and fast.
- Event-driven asynchronous architecture.
- Suitable for real-time features.
- Large middleware ecosystem.
- Easy integration with frontend applications.

3. Database Technology

1. MongoDB Atlas

- Purpose: Cloud-hosted NoSQL database system.

- Justification:

- Flexible document-based schema
- Suitable for dynamic healthcare records
- High scalability
- Cloud-native deployment support
- Faster iterative development

2. Mongoose ODM

- Purpose: Object Data Modeling (ODM) library for MongoDB.

- Justification:

- Simplifies MongoDB database operations
- Provides schema validation
- Improves query handling
- Supports middleware and model abstraction
- Improves maintainability of backend services



4. Authentication and Security Technologies

1. JWT (JSON Web Tokens)

- Purpose: Session authentication and authorization.

- Justification:

- Stateless authentication
- Secure API authorization
- Scalable session handling
- Suitable for role-based access control

2. bcrypt

- Purpose: Password hashing and encryption.

- Justification:

- Secure password storage
- Industry-standard password protection
- Reduces credential exposure risks

3. TLS 1.3

- Purpose: Secure client-server communication.

- Justification:

- Encrypts data transmission
- Prevents interception attacks
- Secures REST API communication

5. Third-Party Service Selection

1. PayHere Gateway

- Purpose: Online payment processing.

- Justification:

- Local Sri Lankan payment support
- Easy integration for MERN applications
- Supports multiple payment methods



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2. Twilio SMS API

- Purpose: SMS notifications and alerts.

- Justification:

- Reliable SMS delivery
- Appointment reminders
- Emergency notifications

3. Nodemailer

- Purpose:

- Justification:

- Lightweight email integration
- Appointment confirmations
- OTP verification emails
- Password recovery support

4. Jitsi Meet API

- Purpose: Telemedicine video consultation support.

- Justification:

- Open-source video conferencing
- Lightweight integration
- Minimal infrastructure overhead
- Suitable for MVP telemedicine implementation

5. Google Maps API

- Purpose: Location and emergency mapping services.

- Justification

- Accurate clinic location mapping
- Consent-based emergency location sharing
- Easy frontend integration



6. Infrastructure and CI/CD

1. Frontend Hosting

- Selected Platforms:

- Vercel
- Netlify

- Justification:

- Robust free-tier support
- Automatic SSL provisioning using TLS 1.3
- Seamless GitHub repository integration
- Continuous deployment pipelines
- Fast global CDN delivery
- Simplified React SPA deployment

2. Backend Hosting

- Selected Platforms:

- Render
- Railway

- Justification:

- Free-tier Node.js hosting support
- Simplified cloud deployment
- Easy GitHub integration
- Automatic deployment workflows
- Environment variable management
- Scalable backend hosting capabilities

3. Version Control

- Selected Tools:

- GitHub
- GitHub Actions

- Purpose:

- Source code management
- Branch protection
- Pull request workflows



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- Sprint issue tracking
- Team collaboration

- Justification:

- Code consistency
- Deployment reliability
- Team collaboration
- Faster issue detection
- Maintainability

3. Architectural Style Justification

SithCare adopts a Layered MVC Architecture with modular service-oriented design principles.

The architecture consists of:

1. Presentation Layer
2. Application / Controller Layer
3. Business Logic Layer
4. Data Access Layer

This architectural style was selected due to the following reasons:

Architectural Benefit	Justification
Maintainability	Independent layers simplify debugging and updates
Scalability	Modular services support future system expansion
Security	Centralized authentication and validation
Flexibility	External APIs can be independently managed
Testability	Layers can be independently tested
Extensibility	Easier future migration toward microservices



3. Conclusion

The selected MERN technology stack and architectural decisions provide a scalable, secure, maintainable, and practical foundation for the SithCare platform. The architecture balances academic feasibility with enterprise-level software engineering principles while supporting future system evolution and healthcare scalability.